

Creation Date 07-Jul-2009

Revision Date 05-Feb-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                          |
|---------------------------|------------------------------------------|
| Product Description:      | <u>Manganese(II) sulfate monohydrate</u> |
| Cat No. :                 | <b>33341</b>                             |
| Synonyms                  | Manganous sulfate monohydrate            |
| CAS No                    | 10034-96-5                               |
| Molecular Formula         | Mn O4 S . H2 O                           |
| REACH registration number | -                                        |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

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## 2.1. Classification of the substance or mixture

### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (repeated exposure)

Category 1 (H318)  
Category 2 (H373)

#### Environmental hazards

Chronic aquatic toxicity

Category 2 (H411)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure if inhaled

H411 - Toxic to aquatic life with long lasting effects

### Precautionary Statements

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P310 - Immediately call a POISON CENTER or doctor/physician

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component                     | CAS No     | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|-------------------------------|------------|-----------|----------|---------------------------------------------------|
| Manganese sulfate monohydrate | 10034-96-5 | 600-072-9 | >95      | Eye Dam. 1 (H318)                                 |

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|                       |           |                   |   |                                                                   |
|-----------------------|-----------|-------------------|---|-------------------------------------------------------------------|
|                       |           |                   |   | STOT RE 2 (H373)<br>Aquatic Chronic 2 (H411)                      |
| Manganese(II) sulfate | 7785-87-7 | EEC No. 232-089-9 | - | Eye Dam. 1 (H318)<br>STOT RE 2 (H373)<br>Aquatic Chronic 2 (H411) |

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                           |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. Causes severe eye damage.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### Hazardous Combustion Products

Sulfur oxides, Manganese oxides.

### 5.3. Advice for firefighters

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As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Protect from moisture.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 13

**Switzerland - Storage of hazardous substances**

Storage class - SC 11/13  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### **Exposure limits**

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

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| Component                     | European Union                   | The United Kingdom                                                                                                                               | France | Belgium | Spain                                                                                           |
|-------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|-------------------------------------------------------------------------------------------------|
| Manganese sulfate monohydrate | TWA: 0.05 mg/m <sup>3</sup> (8h) | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr |        |         | TWA / VLA-ED: 0.2 mg/m <sup>3</sup> (8 horas)<br>TWA / VLA-ED: 0.05 mg/m <sup>3</sup> (8 horas) |
| Manganese(II) sulfate         | TWA: 0.05 mg/m <sup>3</sup> (8h) | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr |        |         | TWA / VLA-ED: 0.2 mg/m <sup>3</sup> (8 horas)<br>TWA / VLA-ED: 0.05 mg/m <sup>3</sup> (8 horas) |

| Component                     | Italy | Germany                                                                                                                                                                                                                                                                                                     | Portugal                                                                  | The Netherlands | Finland                                |
|-------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------|----------------------------------------|
| Manganese sulfate monohydrate |       | TWA: 0.2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.2 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 1.6 mg/m <sup>3</sup><br>Höhepunkt: 0.16 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> 8 horas<br>TWA: 0.05 mg/m <sup>3</sup> 8 horas |                 | TWA: 0.02 mg/m <sup>3</sup> 8 tunteina |
| Manganese(II) sulfate         |       | TWA: 0.2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.2 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 1.6 mg/m <sup>3</sup><br>Höhepunkt: 0.16 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> 8 horas<br>TWA: 0.05 mg/m <sup>3</sup> 8 horas |                 | TWA: 0.02 mg/m <sup>3</sup> 8 tunteina |

| Component                     | Austria                                                                                | Denmark | Switzerland                          | Poland | Norway                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------|---------|--------------------------------------|--------|---------------------------------------------------------------------------|
| Manganese sulfate monohydrate | MAK-KZGW: 1.6 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.2 mg/m <sup>3</sup> 8 Stunden |         | TWA: 0.5 mg/m <sup>3</sup> 8 Stunden |        | TWA: 0.2 mg/m <sup>3</sup> 8 timer<br>TWA: 0.05 mg/m <sup>3</sup> 8 timer |
| Manganese(II) sulfate         | MAK-KZGW: 1.6 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.2 mg/m <sup>3</sup> 8 Stunden |         | TWA: 0.5 mg/m <sup>3</sup> 8 Stunden |        | TWA: 0.2 mg/m <sup>3</sup> 8 timer<br>TWA: 0.05 mg/m <sup>3</sup> 8 timer |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers

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## Predicted No Effect Concentration (PNEC)

See values below.

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

#### Physical State

Powder Solid

#### Appearance

Off-white

#### Odor

No information available

#### Odor Threshold

No data available

#### Melting Point/Range

700 °C / 1292 °F

#### Softening Point

No data available

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Boiling Point/Range                     | 850 °C / 1562 °F         |                                   |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | Not applicable           |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | 3 - 3.5                  | 5% aq. sol                        |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | Soluble                  |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

## 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Mn O4 S . H2 O         |
| Molecular Weight  | 169.02                 |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions. Hygroscopic.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Avoid dust formation. Incompatible products. Excess heat. Exposure to moist air or water.

### 10.5. Incompatible materials

Reducing Agent. Peroxides. Organic materials. Strong bases. Oxidizing agent.

### 10.6. Hazardous decomposition products

Sulfur oxides. Manganese oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

| Component             | LD50 Oral                | LD50 Dermal | LC50 Inhalation              |
|-----------------------|--------------------------|-------------|------------------------------|
| Manganese(II) sulfate | LD50 = 782 mg/kg ( Rat ) | -           | LC50 > 4.45 mg/L ( Rat ) 4 h |

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|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

|                                                               |                                                                                                                                      |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| (b) skin corrosion/irritation;                                | Based on available data, the classification criteria are not met                                                                     |
| (c) serious eye damage/irritation;                            | Category 1                                                                                                                           |
| (d) respiratory or skin sensitization;<br>Respiratory<br>Skin | Based on available data, the classification criteria are not met<br>Based on available data, the classification criteria are not met |
| (e) germ cell mutagenicity;                                   | Based on available data, the classification criteria are not met                                                                     |
| (f) carcinogenicity;                                          | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product        |
| (g) reproductive toxicity;                                    | Based on available data, the classification criteria are not met                                                                     |
| (h) STOT-single exposure;                                     | Based on available data, the classification criteria are not met                                                                     |
| (i) STOT-repeated exposure;<br>Target Organs                  | Category 2<br>Central nervous system (CNS), Liver, Kidney, Blood, Respiratory system.                                                |
| (j) aspiration hazard;                                        | Not applicable<br>Solid                                                                                                              |
| Symptoms / effects, both acute and delayed                    | No information available.                                                                                                            |

## 11.2. Information on other hazards

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Endocrine Disrupting Properties | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

|                     |                                                                                                                                                                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects | Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system. |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                     |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>12.2. Persistence and degradability</u><br>Persistence<br>Degradability<br>Degradation in sewage treatment plant | Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary based on information available, May persist.<br>Not relevant for inorganic substances.<br>Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                        |                                          |
|----------------------------------------|------------------------------------------|
| <u>12.3. Bioaccumulative potential</u> | May have some potential to bioaccumulate |
|----------------------------------------|------------------------------------------|



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## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors.

## 12.7. Other adverse effects

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance.  
This product does not contain any known or suspected substance.

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN3077

#### 14.2. UN proper shipping name Technical Shipping Name

Environmentally hazardous substances, solid, n.o.s.  
Manganese (II) sulfate

#### 14.3. Transport hazard class(es)

9

#### 14.4. Packing group

III

### ADR

#### 14.1. UN number

UN3077

#### 14.2. UN proper shipping name Technical Shipping Name

Environmentally hazardous substances, solid, n.o.s.  
Manganese (II) sulfate

#### 14.3. Transport hazard class(es)

9

#### 14.4. Packing group

III

### IATA

#### 14.1. UN number

UN3077

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|                                                                      |                                                                                                          |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.2. UN proper shipping name</b>                                 | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.*                                                      |
| <b>Technical Shipping Name</b>                                       | Manganese (II) sulfate                                                                                   |
| <b>14.3. Transport hazard class(es)</b>                              | 9                                                                                                        |
| <b>14.4. Packing group</b>                                           | III                                                                                                      |
| <b>14.5. Environmental hazards</b>                                   | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
| <b>14.6. Special precautions for user</b>                            | No special precautions required.                                                                         |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods                                                                           |

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                     | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-------------------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Manganese sulfate monohydrate | 10034-96-5 | -         | -      | -   | X     | X    | -        | X    | X    |
| Manganese(II) sulfate         | 7785-87-7  | 232-089-9 | -      | -   | X     | X    | KE-23032 | X    | X    |

| Component                     | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | AICS | NZIoC | PICCS |
|-------------------------------|------------|------|-----------------------------------------------|-----|-----|------|-------|-------|
| Manganese sulfate monohydrate | 10034-96-5 | -    | -                                             | -   | -   | X    | X     | X     |
| Manganese(II) sulfate         | 7785-87-7  | X    | ACTIVE                                        | X   | -   | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

Not applicable

| Component                     | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Manganese sulfate monohydrate | 10034-96-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Manganese(II) sulfate         | 7785-87-7  | -                                                                   | -                                                                             | -                                                                                                     |

#### Seveso III Directive (2012/18/EC)

| Component                     | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Manganese sulfate monohydrate | 10034-96-5 | Not applicable                                                                            | Not applicable                                                                           |
| Manganese(II) sulfate         | 7785-87-7  | Not applicable                                                                            | Not applicable                                                                           |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

#### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at

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work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component             | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------------|---------------------------------------|-------------------------|
| Manganese(II) sulfate | WGK2                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

H411 - Toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Training Advice

Chemical incident response training.

Prepared By  
Creation Date

Health, Safety and Environmental Department  
07-Jul-2009

# SAFETY DATA SHEET

Manganese(II) sulfate monohydrate

Revision Date 05-Feb-2024

Revision Date 05-Feb-2024

Revision Summary New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**